

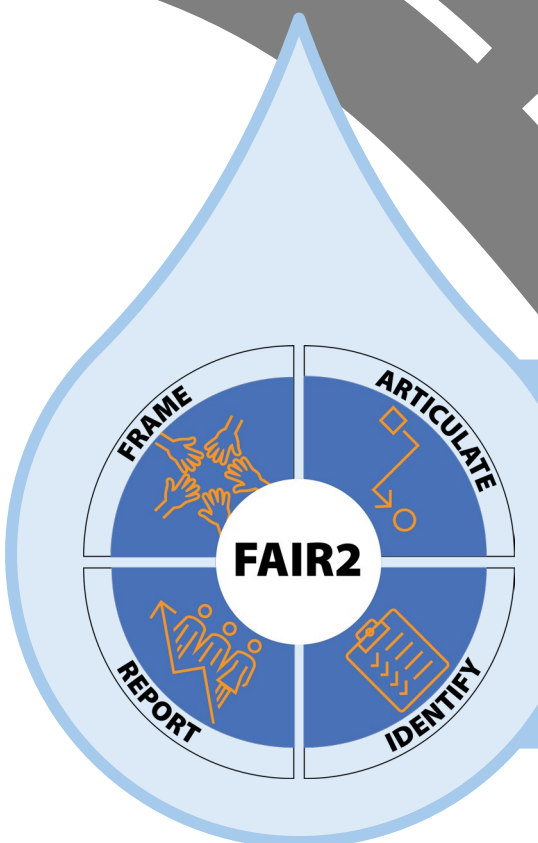
Demographic Data in Integrated Administrative Data Systems:

**Seeking Experiential Knowledge to
Inform Race and Gender Data**

Module 2 of 5

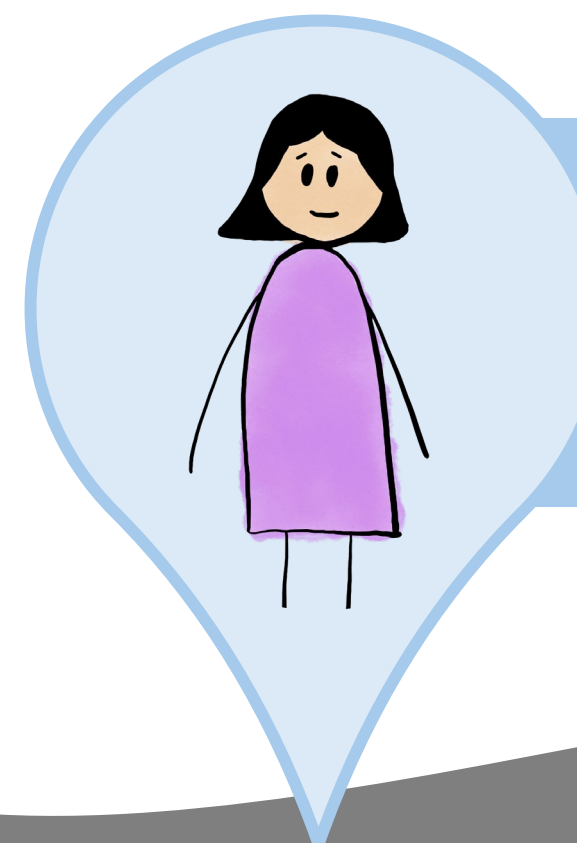
**We want to learn
from your
experience!**

Public Interest Technology University Network Year 6 Challenge
January 2026

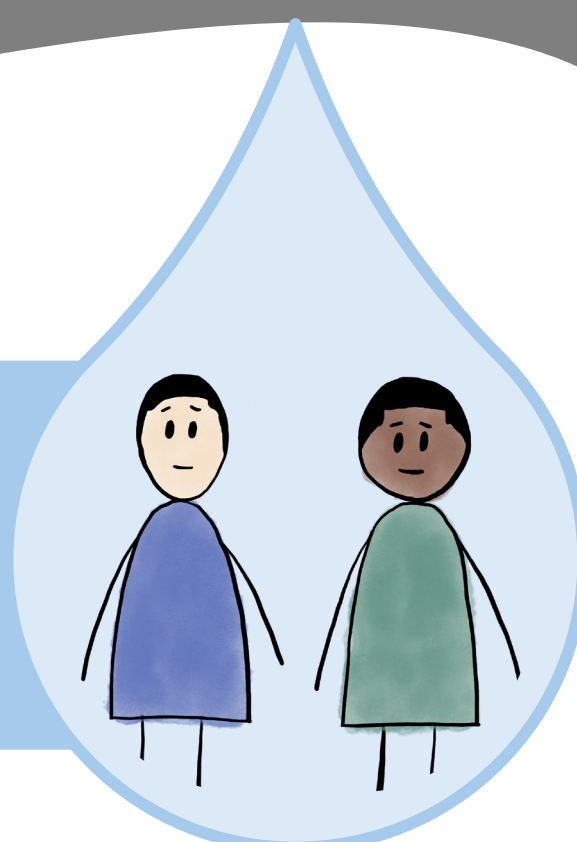


Module 1
FAIR2: A framework for integrating community knowledge in data analytics and addressing discrimination bias

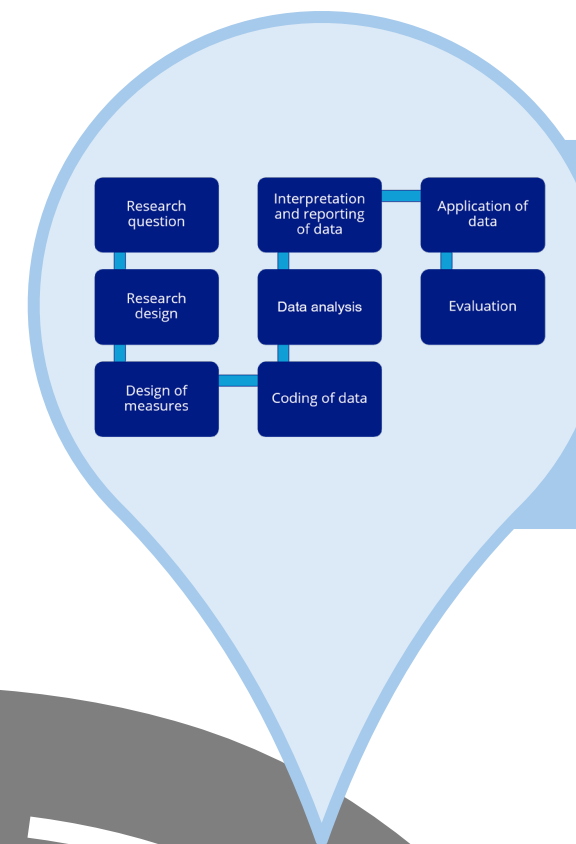
Module 2
Demographic Data in Integrated Data Systems: Seeking Experiential Knowledge to Inform Race and Gender Data



Module 3
An illustration of Label Bias in Homelessness Data Analytics

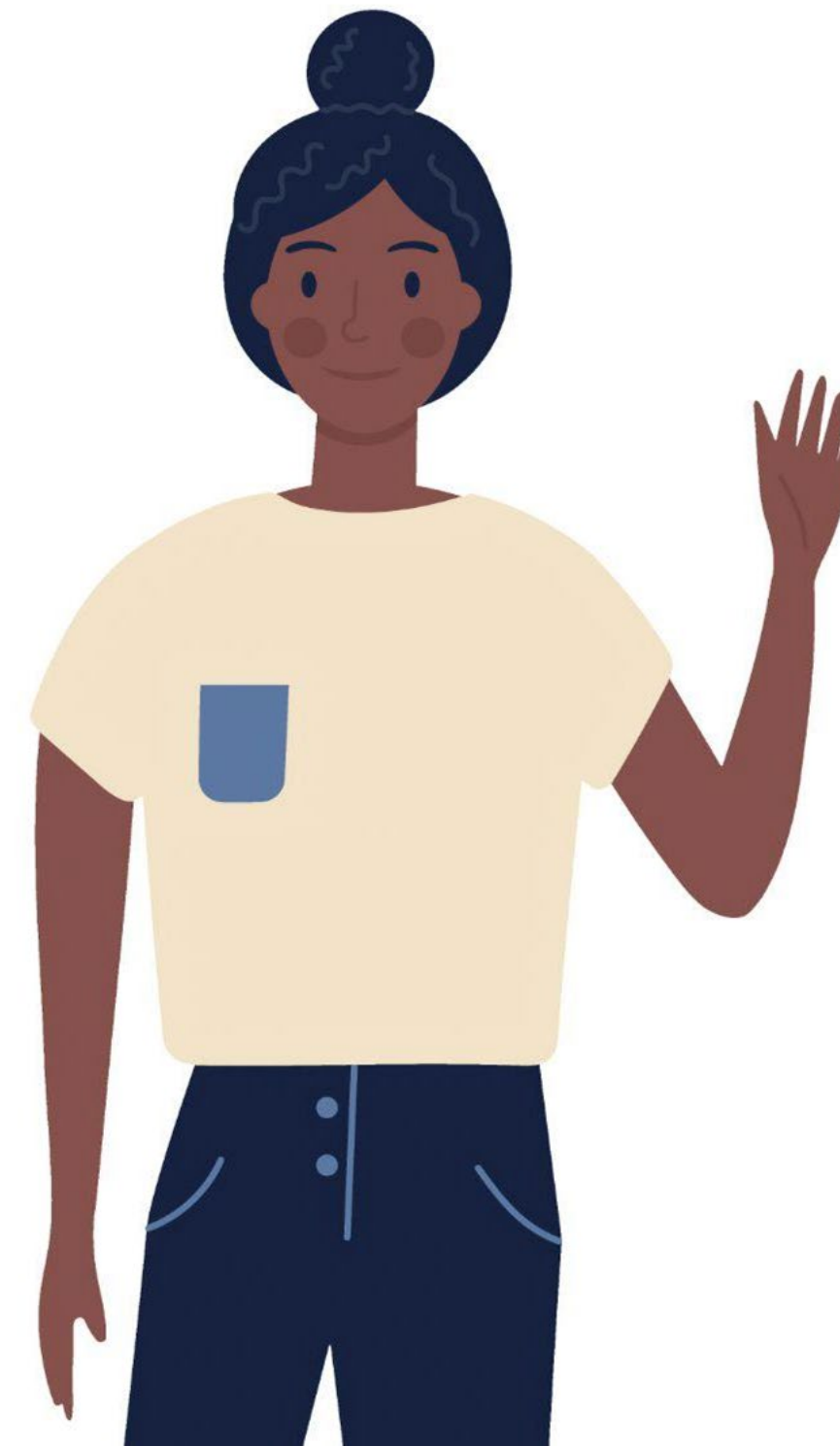


Module 4
An illustration of Collider Bias in housing research



Module 5
Recommendations to address constraints and bias in Integrated Data Analytics

Meet Aaron



2005

Birth



HMIS - Family Profile

Race	Black
Ethnicity	Unknown
Gender	Male



SNAP - Family Profile

Race	Black
Ethnicity	Non-Hispanic
Gender	Male



HMIS - Family Profile

Race	Multi-Racial
Ethnicity	Non-Hispanic
Gender	Male



SNAP - Family Profile

Race	Multi-Racial
Ethnicity	Non-Hispanic
Gender	Male

Aaron doesn't know much about their early years, though they know their parents had them very young and struggled to make ends meet.

Neither of Aaron's parents had any family that lived nearby, so Aaron knows that they sometimes got help from food pantries and federal programs, like SNAP.



2011

1st Grade



School - Elementary

Race / Ethnicity	Multi-Racial
Gender	Male



SNAP - Family Profile

Race	Multi-Racial
Ethnicity	Non-Hispanic
Gender	Male



School - Middle School

Race / Ethnicity	Black
Gender	Male

2020

9th Grade

Throughout elementary school and middle school, Aaron's family moved around a lot based on where their parents could find work. Aaron remembers that sometimes money would get tight around the end of the month, but their parents always found a way to get by.



2020

9th Grade



SNAP - Family Profile

Race	Multi-Racial
Ethnicity	Non-Hispanic
Gender	Male



School - High School

Race / Ethnicity	Black
Gender	Other

2023

12th Grade

Aaron is very happy that their family was able to spend all four years of high school in the same city, though. They finally felt like they fit in after making friends with a group of other LGBTQ+ students and started to explore both their gender identity and racial identity. Having parents from two different racial backgrounds, Aaron was never quite sure where they fit in. Aaron's high school friends helped them form their own understanding of their identities. After high school, Aaron's parents moved again, but Aaron decided to stay in the same city that they now called home.



2024
HS Graduation



SNAP - Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Unknown



SNAP - Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Unknown



HMIS – Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Non-Binary



HMIS – Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Male

2025
Present

Aaron knew they wanted to get a college education but wanted to earn some income as well. So, they immediately started a part-time job at a local restaurant while applying for college. It wasn't enough money to make ends meet, so Aaron found themselves applying for Supplemental Nutrition Assistance Program (SNAP) services. With SNAP and their part-time job, Aaron was able to rent a small apartment and take a couple college classes up until a few weeks ago. Aaron had been growing more confident in their gender expression and identity, but unfortunately that brought on multiple ongoing customer complaints at the restaurant they worked at. Unfortunately, Aaron's manager thought it would be easier to let Aaron go and appease the customers, so Aaron suddenly became unemployed with no way to pay next month's rent.



2024
HS Graduation



SNAP - Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Unknown



SNAP - Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Unknown



HMIS – Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Non-Binary

2025
Present



HMIS – Adult Profile

Race Black
Ethnicity Non-Hispanic
Gender Male

Aaron has applied for some housing assistance programs but will soon lose the apartment. Aaron has been looking into homeless shelters in the area but is concerned they won't feel safe at the men's shelter and won't be allowed in the women's shelter. On top of all of that, Aaron never got around to changing their ID and documents from "male" and isn't sure what to do about all the forms they now have to fill out for housing assistance.

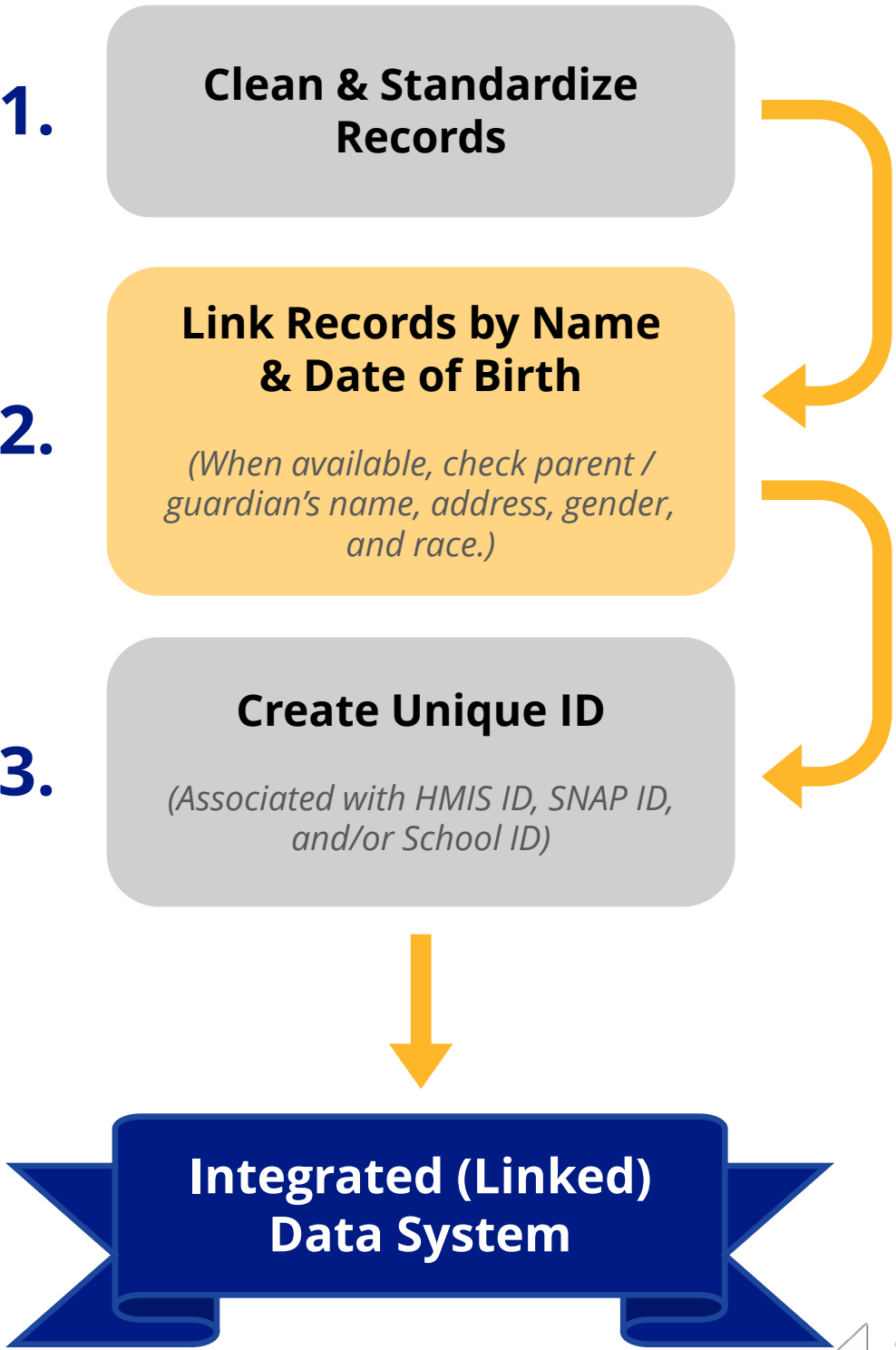


What is Integrated Administrative Data (IDS)?

Linked administrative data is created when information about the same person from different public systems, such as homelessness services (HMIS), food assistance (SNAP), and schools, is:

- 1. Cleaned
- 2. Matched using shared info like name and date of birth
- 3. Connected via a unique ID

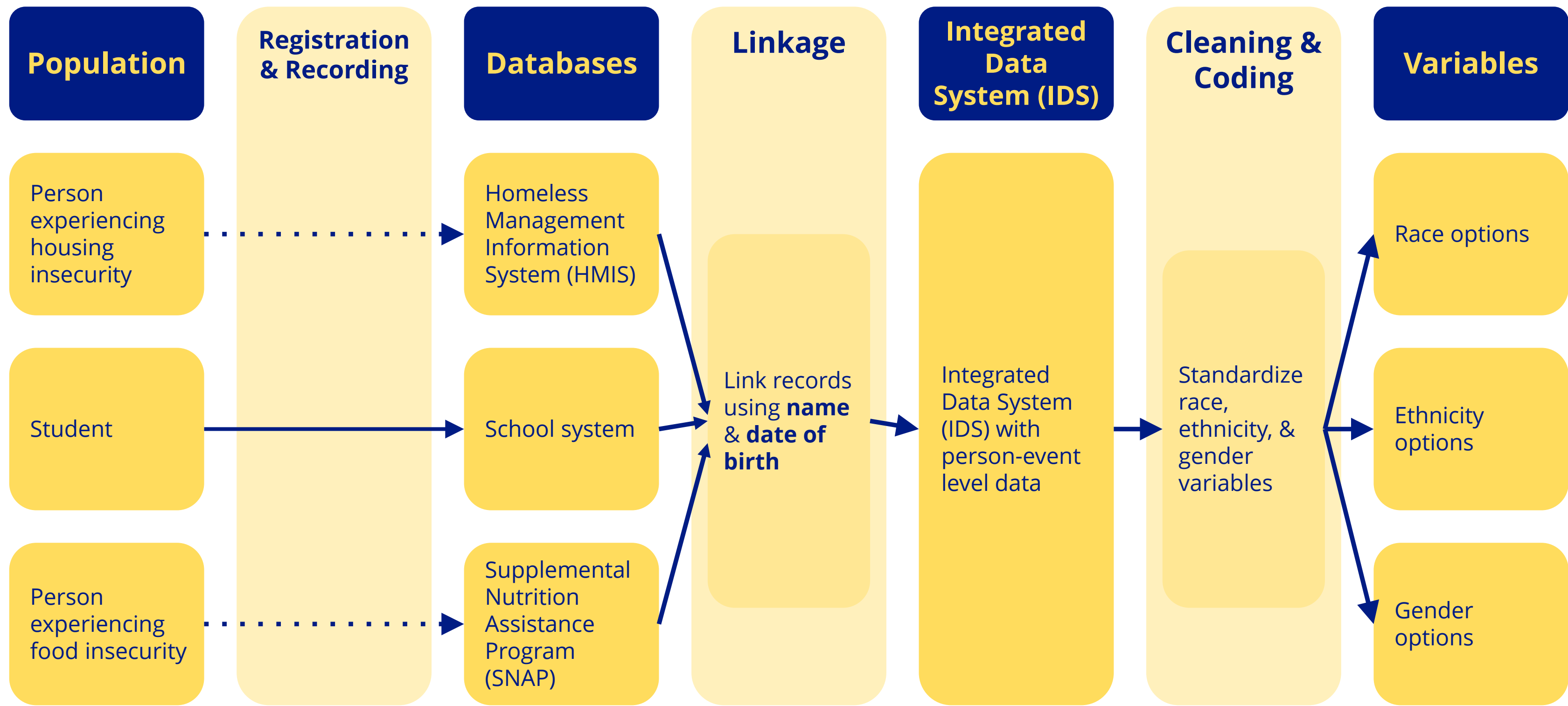
Individual Data Sources			
	HMIS	SNAP	School
Name	✓	✓	✓
Date of Birth	✓	✓	✓
ID	HMIS ID	Snap ID	School ID
Race	HMIS-Defined Race Category	SNAP-Defined Race Category	School Race / Ethnicity Code
Ethnicity	HMIS-Defined Ethnicity Category	SNAP-Defined Ethnicity Category	X
Gender	HMIS-Defined Gender Category	SNAP-Defined Gender Category	School-Defined Gender
Time Frame	Date of Service	Month of Enrollment	Year and Grade
System-Specific Data	Type of Service	Benefit Amount	Homeless Status
Location	Previous Housing Status	Current Address	Current Address



Learning from Linked Administrative Data and Experiential Knowledge

- Integrated Data Systems (IDS) allows us to see how individuals interact with multiple programs over time
- In a Secured Research Environment, under strict research ethics protocol and data use agreements, we **link data for individuals** interacting with homeless services
- We analyze this data to learn about the **needs and assets in our community** and inform positive social change
- **Data do not speak for themselves**
 - We can learn more about what data represents by tapping on the collective expertise of those who sought out homeless services and provided information that makes up the administrative data

Linking Administrative Data to Create an Integrated Data System (IDS)



Eliciting Experiential Knowledge: Data Chats to Learn About How Data is Generated

- **Data Chat:**
 - A method of leveraging experiential knowledge where researcher and people with lived experience share and discuss data
 - Rooted in the FAIR2 framework for identifying bias in social data science (Richter et al., 2023)
- We held two Data Chats in **Cleveland, OH**, on **demographic data collection in homeless services** and other administrative systems
 - Participants were recruited through flyers at community organizations
 - Participants were **young adults with lived experience of homelessness**
 - We share the data discussed followed by participants' insights



What we shared with Data Chat participants

- **Summary statistics** of data linked across three systems:
 - Homeless Management Information system (HMIS)
 - Supplemental Nutrition Assistance Program (SNAP)
 - Cleveland Metropolitan School District (CMSD)
- Individuals in the data:
 - Ages 18-25 years
 - Those who interacted with homeless services in Cuyahoga County
 - 2017, 2021 and 2021-2023
- When **linking records of individuals**, how do we cope with **different classifications of race and gender id** across agencies and over time?

Demographic information is collected in administrative data in different formats and ways across systems

HMIS		
Entry Year	Race	Percent
2021	American Indian, Alaska Native, or Indigenous	0.2%
2021	Asian or Asian American	0.3%
2021	Black, African American, or African	76.9%
2021	Client refused	0.6%
2021	Mutli-Racial	3.8%
2021	Native Hawaiian or Pacific Islander	0.1%
2021	White	18.1%

CMSD		
School Year	Race/Ethnicity	Percent
2021	Asian	1.1%
2021	Black	68.3%
2021	Hispanic	0.9%
2021	Native American	0.4%
2021	Multiracial	0.3%
2021	Pacific Islander	0.4%
2021	Unknown/Not applicable	0.2%
2021	White	28.4%

SNAP			
Year	Race	Count	Percent
2021	American Indian or Alaskan Native	8,191	1.3%
2021	Asian	7,945	1.3%
2021	Black or African American	413,214	67.4%
2021	Missing/unknown	18,848	3.1%
2021	Native Hawaiian or Other Pacific Islander	1,813	0.3%
2021	White	163,355	26.6%



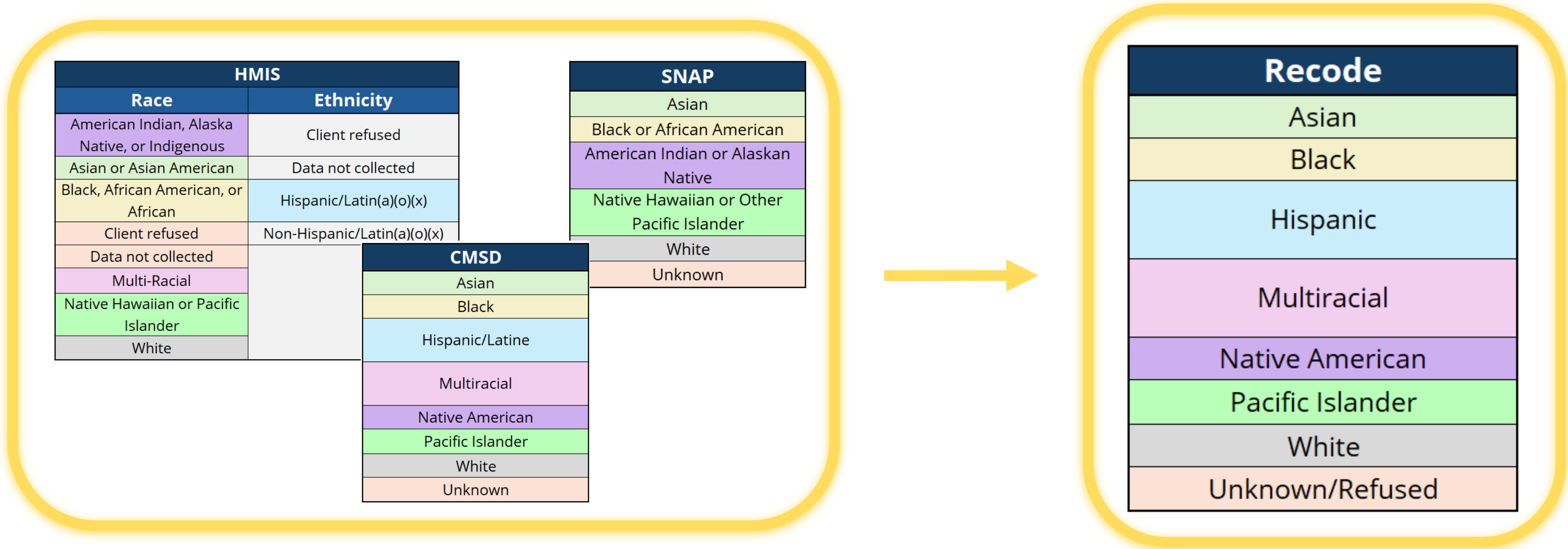
Sometimes classifications change over time within the same agency

HMIS		
Year	Gender	Percent
2017	Female	48.6%
2017	Male	50.5%
2017	Trans Female	0.3%
2017	Trans Male	0.0%
2017	Transgender	0.6%
2021	A gender other than singularly female or male	0.0%
2021	Female	52.5%
2021	Male	46.8%
2021	Questioning	0.0%
2021	Transgender	0.6%

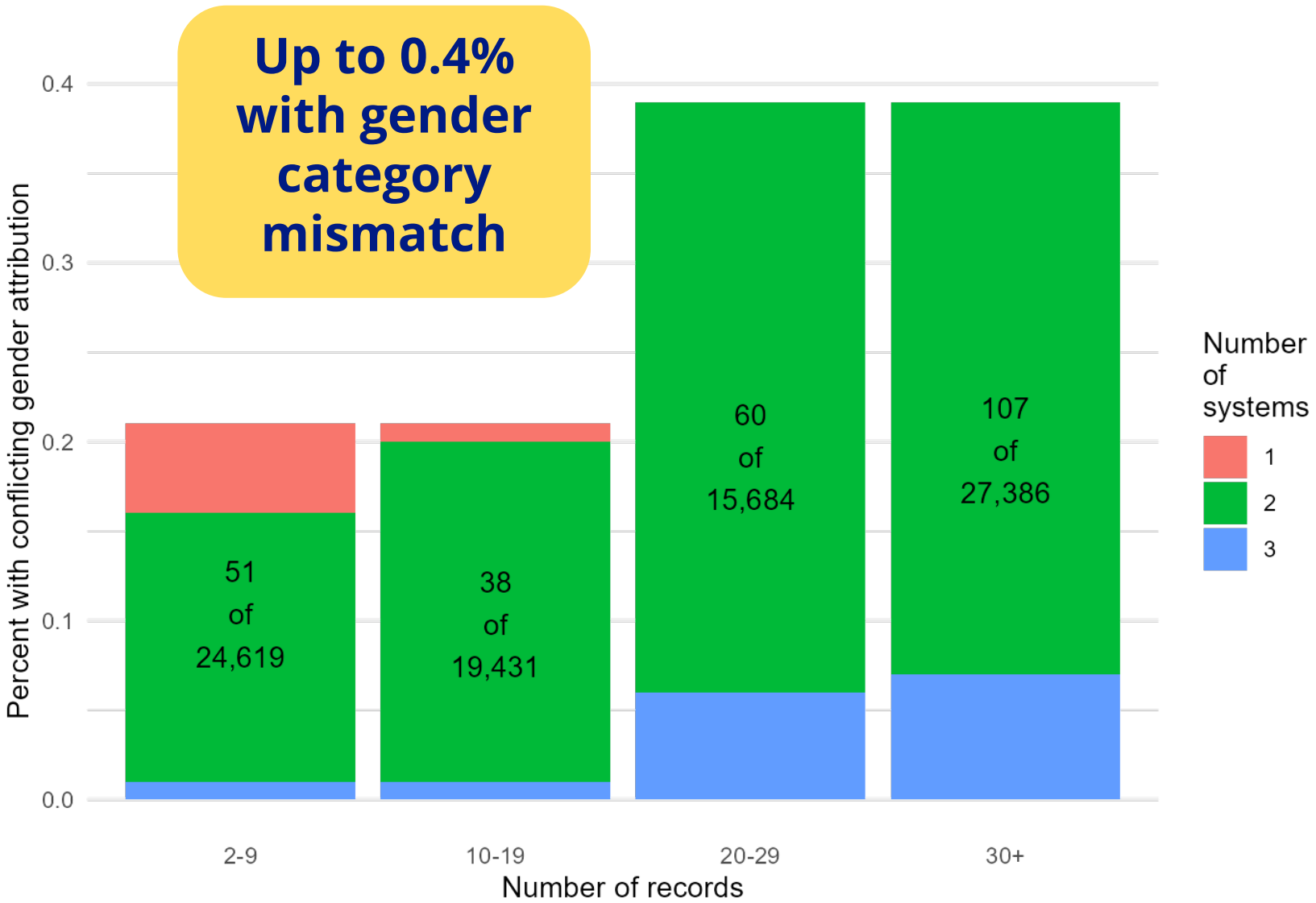
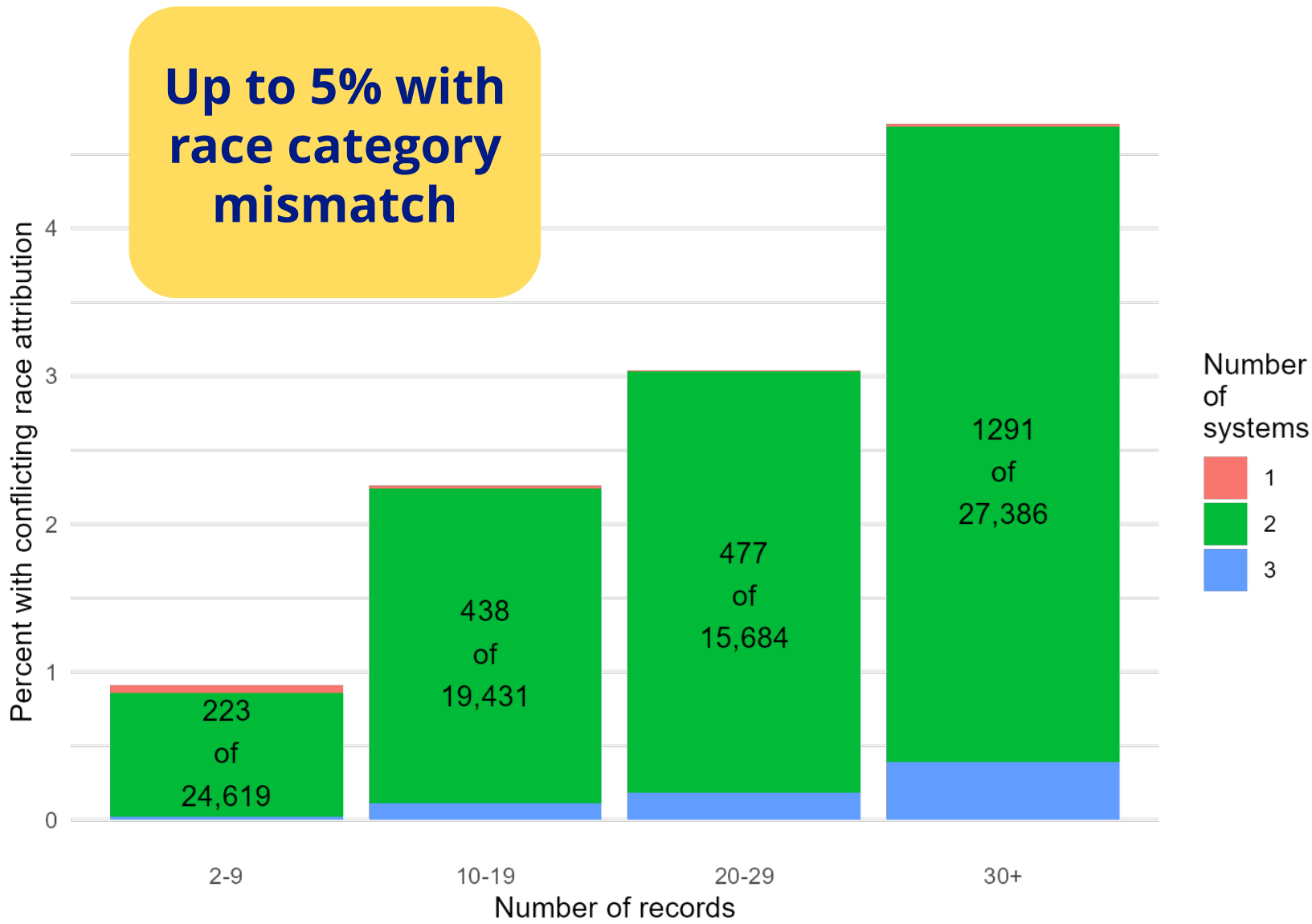


Mapping different demographic categories into standardized, inclusive, common values

- We map each agency's classification into a **standardized common categorization** that **minimizes the loss of inclusive options** in the linked data to **maximize diverse identities**



Even after standardizing categories, for some people, demographic information changes across multiple administrative records



Black/White/Multiracial comprise the majority of cases where race data reporting changes for the same person

Female/male and male/transgender are most common patterns of gender reporting that changes for the same person

These discrepancies arise for reasons that DC participants helped us uncover or contextualize

1. Agency-defined demographic categories:

- Demographic fields in administrative data are **not inherent or fixed characteristics** of individuals
- Demographic fields represent agency-defined classification schemes rooted in social norms and history that may differ across agencies or change over time

"That was me. I changed my name" ... they kind of give me like a hard time, because then they'd be like. 'Well, this doesn't match up.'"

"My personal feelings ... is that the only thing that needs to be identified is 'Yes, I'm human.'"

2. Changes in names due to adoption or identity transition:

- Records of people who changed their name due to a change in gender id or adoption will not be linked
- These records may be counted as belonging to two different individuals, leading to an undercount representation of trans or gender diverse individuals in the linked data

"I feel like I am my own, an entity, and that is good enough for me. So, I never see myself represented on there, and I'm never going to, because they don't put [participant's name] ... that's not an identity, right?"



3. The Data Intake Experience: Data reporting will be impacted by...

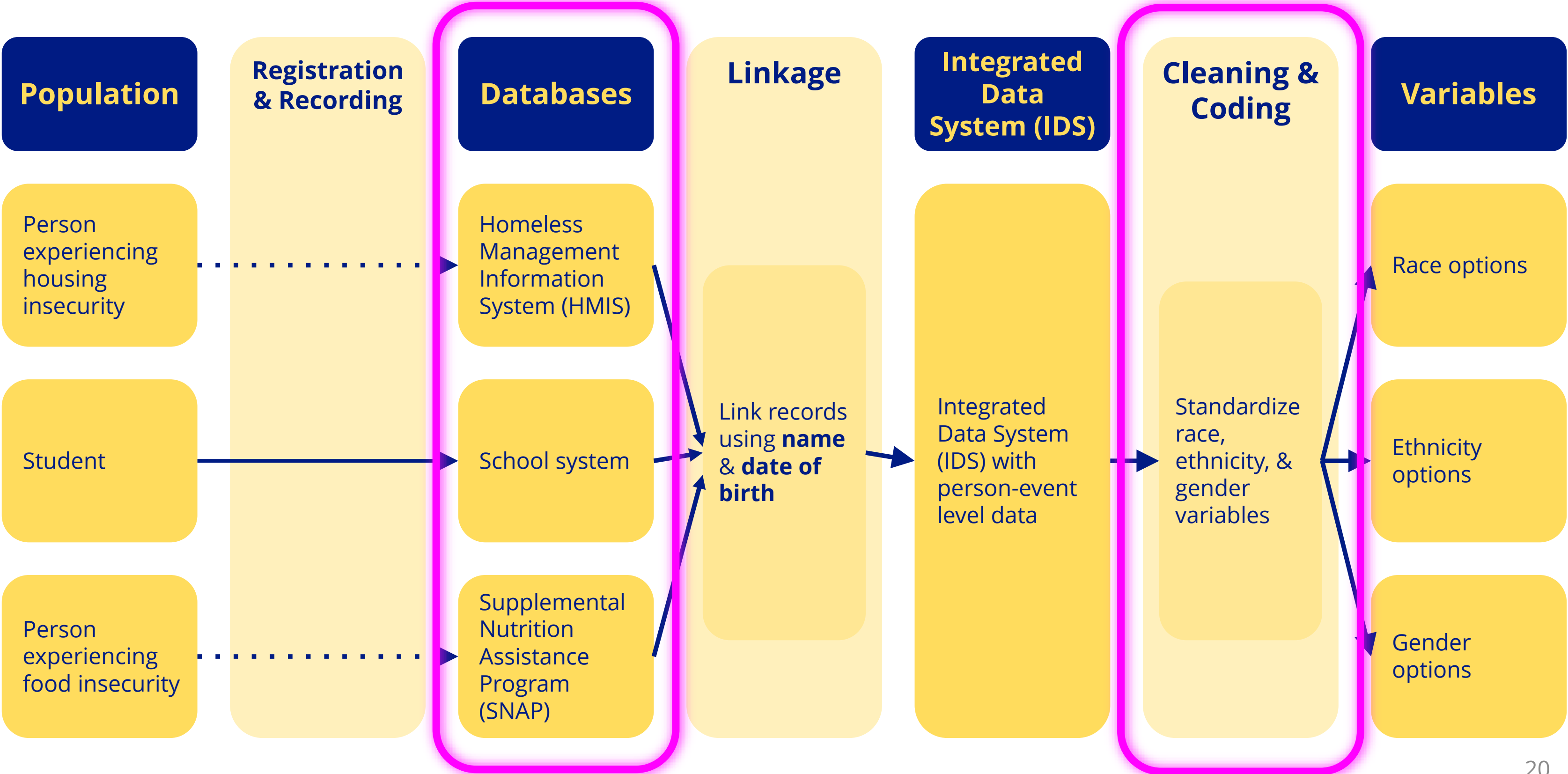
- What, when and how data is collected
- Concerns for **safety, stigmatization, service needs**
- **Burden of data collection**
- We unpack these points with insights and recommendations from participants to ease the burden of data collection and use data in ways that uplifts everyone and promotes community engagement

“The worst thing is that you share your information with somebody and instead of them helping, they’re judging and categorizing you... And this is the person that’s going to help me, so you’re relying on them to really be trustworthy.”

“Being on the street, we’re not thinking about the color we are, so why is the government thinking about what color we are?”



Human choices shape race and gender data at **two points**: when agencies define the categories on their forms, and when researchers standardize those categories to link records across systems.



Data Chat Participants' Insights



- Given your experience with demographic data collection, **what would you suggest to improve the data collection process?**

Discussion



Data Collection Process

Need for Inclusive & Standardized Options

- Ex. Some forms say "sex," some say "gender"
- Not all identities are represented in options

Standardize Data Collection Process

- Self-report vs. provider asking questions

Improve Training for Providers

- Training on how to ask sensitive questions
- Reduce implicit bias



Data Intake Experience



- Are there situations in the data intake process for homeless services that may cause you to **not** want to answer to questions about race identity, ethnicity, and/or gender (**disclosure of personal information**)?

Discussion



Disclosure of Personal Information

Influenced by Safety & Stigma

- Decision to disclose personal information (particularly for LGBTQ+ folks) is influenced by safety
- Particularly a concern for trans women
- *"If you're not a cisgender person, **you are looked at as the most dangerous person, but in my eyes, a cisgender person is the most dangerous person to me.**"*

Affects Receipt of Services

- May not disclose for concern of not receiving adequate services
- Oftentimes, documents (i.e., gender marker, name) have to match to receive services

Influenced by Level and Urgency of Need

- If someone's needs are high or urgent, they may choose what to disclose based on what they think will get them help fastest



Data Intake Experience



- **How important is it** to collect these demographic variables? **Why?**
- How does the data intake experience **address your needs?** Where does it fall short?

Discussion



Necessity of Data Collection

Important for Funding

- Funding requires organizations to report on who they are servicing
- Funds may be **available for certain populations**, so it's important to know what populations are serviced

Important in Identifying Gaps

- Collecting demographic data can help identify care gaps, which populations are not serviced equally and equitably
- *"[Race] still plays a huge part. History definitely ties into that because if you were given less from the beginning and then there's people that were given more, then **you're constantly trying to play catch up in a system that wants you to stay down.**"*



Necessity of Data Collection (Con't)

Purpose Matters

- Could be beneficial if used for **positive purposes** (i.e., making sure you're in a safe environment, **LGBTQ+ inclusive** housing)
- People want to know why you're asking, otherwise it could be a **safety concern**
- Other metrics (**support, resources**) could be more important than demographics
- *"When they do the questionnaires, they don't ask your sexuality, so why is race played into seeking homeless services? **Race shouldn't really be a factor in someone needing shelter or food in their mouth.**"*
- *"... When I aged out of foster care, they have all these statistics about foster kids who complete college, who do not complete college, and how many of us are going to end up homeless, right? ... **What is the point of putting that burden on someone at 18 who's about to venture off? Saying these are the statistics of someone who is like you, who looks like you.**"*



Your Advice



- We asked Data Chat participants... Based on your experience, what could help **reduce discrepancies in data** for people facing homelessness?



Discrepancies in the Data

Need for Increased Safety

- Safety changes across **time and place**
- Marginalized populations, particularly undocumented immigrants, may not disclose information
- *"It's not safe for them, especially undocumented immigrants. **It's not safe for them at all.**"*

Consider Effect of Social Norms

- Social pressure to adhere to norms
- *"Until very recently, **the option to be [gender] inclusive was not a thing** because our society wasn't ready for that."*
- Binary trans folks may not mark "transgender" especially considering where they are in their medical transition



Discrepancies in the Data (con't)

Need for Administrative Consistency

- **Records not always updated** by providers
- **Name changes** may cause issues for researchers linking data; could result in both **duplication** and **undercounting** (ex. a trans person who changes their name may not be counted in the trans population but may be doubly counted in the binary gender populations)
- *"I don't think the data itself is wrong... we need to do better at getting it more **accurate**."*

Consider Who is Reporting

- Ex. Some children's records may be parent-reported and other records may be that same child who has now grown up and is seeking services



Your Advice



- We asked Data Chat participants... Based on your experience, what could help **reduce the burden** of data intake for people facing homelessness?



Reducing the Burden of Data

You Can't.

- Data intake is **inherently burdensome** on marginalized populations
- *"I'm conflicted because I think my job would want me to believe that it is necessary to get this information. **I couldn't really give two F's about data.**"*

Consider Timing of Data Collection

- Collect data when folks are in a **safe place** (i.e., rehab facility, shelter, soup kitchen)

Provide Resources, Compensation

- Provide resources, compensation, food, transportation, etc.



Reducing the Burden of Data (Con't)

Involve Those with Lived Experience and Shared Identities

- Involve folks with **lived experience**
- People may be more comfortable with service providers with **shared identities**
- *"Maybe in the future, as youth and young adults lead more of this type of work, **my hope is that it does become a lot less burdensome.**"*

Ensure Accessibility

- Ensure services are accessible
- Ex. Literacy considerations

People Behind the Data

- Service providers may lack a relational component
- Clients may feel like "just a number"
- *"[Providing food or transient support on the frontlines] does provide a certain warmth that there is still someone out there that cares, but at the same time, **it's kind of cruel because you know in the back of your head 'I need more help than this. But thank you.'**"*



Data Chat Summary

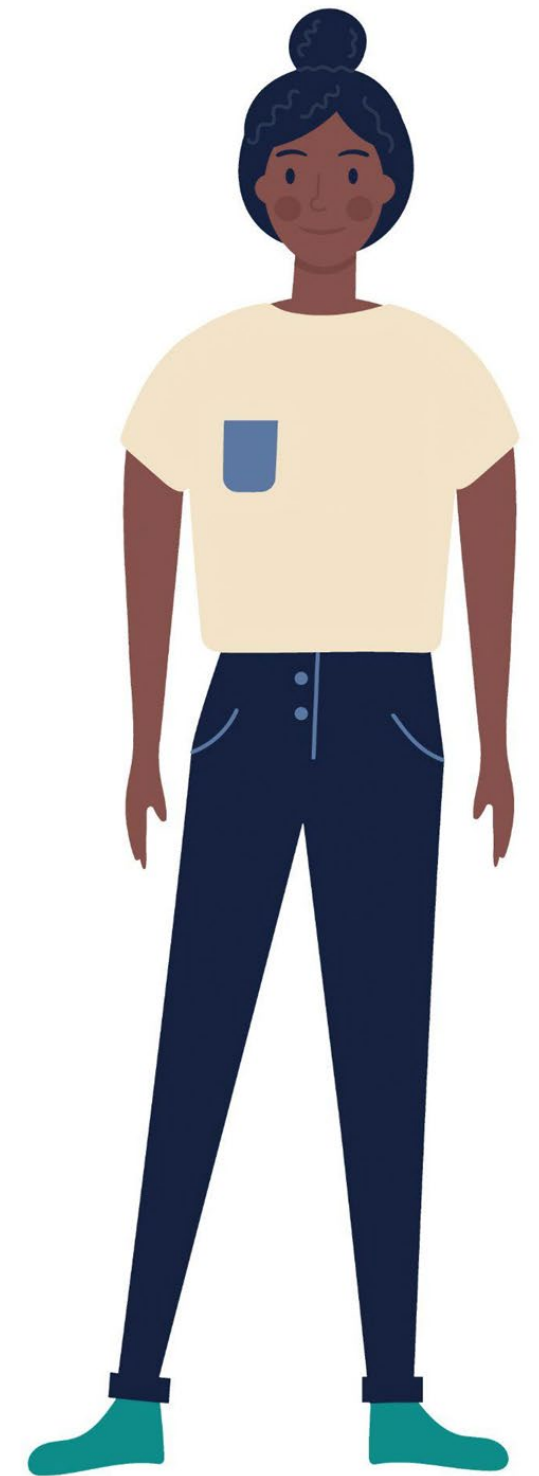
- **Re: Administrative Data Systems, let's advocate for:**
 - More inclusive and standard documentation of people's identities
 - Provider training so that data intake is sensitive to people in vulnerable situations
 - Accessibility and safety as standards in service and data intake
- **When linking administrative data to inform positive social change:**
 - Maximize inclusivity when standardizing categorizations of race and gender identities
 - Consider the role of society in generating the data we analyze
 - Who might be missing in the data due to stigma or safety concerns?
 - Historical context: why are some groups overrepresented in the data?
- **Remember the people who make up the data**
 - Contextualize numbers with **experiences**
 - Data tells **someone's story**
 - Ask for their insights and **report back** to them

*"You're **already in survival mode** to where even being at these places [service providers], you kind of just want to run."*

*"My personal feelings on it is that the only thing that needs to be identified is **'Yes, I'm human.'**"*



- What parts of **Aaron's story** would have been **lost** if we only had the administrative data?
- How can systems **work together** in ways that would have **better supported** Aaron?
- How would Aaron be **represented** if...
 - We only had access to the SNAP data system?
 - We only had access to the HMIS data system?
 - We only had used the information their family reported about Aaron when they were a child versus the information Aaron reported about themselves as an adult?



“I feel like people are eager to share their experience if people are willing to listen.”



References

Shaw, R. J., Harron, K. L., Pescarini, J. M., Pinto Junior, E. P., Allik, M., Siroky, A. N., Campbell, D., Dundas, R., Ichihara, M. Y., Leyland, A. H., Barreto, M. L., & Katikireddi, S. V. (2022). Biases arising from linked administrative data for epidemiological research: a conceptual framework from registration to analyses. *European journal of epidemiology*, 37(12), 1215–1224. <https://doi.org/10.1007/s10654-022-00934-w>

Richter, F.; Nelson, E.; Coury, N.; Bruckman, L.; Knighton, S. (2023). FAIR2: A framework for addressing discrimination bias in social data science. Editorial *Universitat Politècnica de València*. 327-335. <https://doi.org/10.4995/CARMA2023.2023.16400>